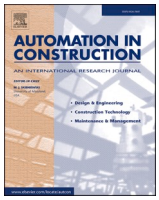
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Review



Natural language processing for smart construction: Current status and future directions

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ABSTRACT

Unstructured texts dominate data in construction projects. With the achievements of natural language processing (NLP) techniques, mining unstructured text data for smart construction has become increasingly significant. To understand state-of-the-art NLP for smart construction, uncover related issues, and propose potential improvements, this paper presents a comprehensive review of bottom-level techniques and mainstream applications of NLP in the industry. In total, 124 journal articles published in the last two decades are reviewed. NLP involves five core steps supported by various techniques, e.g., syntactic parsing, heuristic rules, machine learning, and deep learning. NLP has been applied for information extraction and exchanging and many downstream applications to facilitate management and decision-making. The role of NLP in smart construction and current challenges for fully reaping its benefits are discussed, and four research directions are identified, i.e., improving relation extraction, realising knowledge base auto-development, integrating multi-modal information, and achieving an accuracy-efficiency trade-off by developing an NLP application framework. It is envisioned that outcomes of this paper can assist both researchers and industrial practitioners with appreciating the research and practice frontier of NLP for smart construction and soliciting the latest NLP techniques.

1. Introduction

Human history has experienced three industrial revolutions related to mechanization, electrification, and automation, respectively. The world is witnessing the fourth revolution (i.e., industry 4.0), which can significantly improve the way of manufacturing through ubiquitous connectivity and informed decision-making enabled by big data and artificial intelligence (AI). These technologies have a far-reaching influence on the construction sector (CS) and lead to smart construction (construction 4.0) which by referring to industry 4.0, is regarded as integrated adoption of information and automation technologies in construction projects to address engineering challenges [1].

The essence of smart construction is to 1) generate a digital twin of a project through real-time data collection and integration; 2) simulate all life-cycle activities, i.e., planning, design, construction, and operation and maintenance (O&M); 3) optimise decision-making in the activities; 4) carry out the physical project following optimised decisions, which brings various benefits, e.g., minimising costs, compressing schedules, and increasing quality, safety, and productivity.

Smart construction heavily relies on data to build and maintain the interaction between the physical and virtual world. Data include structured and unstructured data. Structure data (e.g., sensor data) is often stored in column-row tables which however cannot effectively handle unstructured data (e.g., images and texts). Current smart construction focuses on structured data, for instance, information platforms (e.g., Building Information Modelling (BIM), Geographical Information System (GIS), and project systems built on Internet-of-things) rely on table databases [2]. However, over 80% data in CS is unstructured, most of which are texts. Moreover, compared to images/audios, it is much easier and cheaper to collect text data in construction projects [3]. Hence, it is important and cost-effective to extract and understand text data automatically and intelligently.

Natural Language Processing (NLP) can fulfill above purpose, which is defined as techniques which help machines understand human languages through analysing structures of texts and meanings of words. Meanwhile, AI is a broad concept, including all machines/programs that simulate human intelligence using different techniques, e.g., rules, statistic models, and machine learning (ML) and deep learning (DL)

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models. Thus, the relationships between NLP and AI are twofold. First, NLP is a subset of AI and is enabled by AI techniques which are interchangeably called NLP techniques in the paper (more details are presented in Section 4). Second, AI and NLP evolve together. The development of NLP relies on advancement of AI which can enables/improves NLP applications. For example, ML/DL models which can learn common patterns from inputs and outputs and then make judgements autonomously, are trained to replace traditional programs based on hand-crafted rules so that language analysing models are learned automatically rather than programmed manually [4].

NLP has been increasingly adopted in CS, which has four main application scenarios: 1) filtering information, i.e., extracting key information from noisy texts for specific purposes (e.g., finding accident causes from reports), 2) organising documents, i.e., automatically grouping documents of different backgrounds (e.g., drawings of different disciplines) and enabling timely retrieval, 3) expert systems, i.e., integrating expert knowledge and provide answers for engineering problems, and 4) automated compliance checking, i.e., automatically comparing as-is situation (e.g., working plans) with requirements (e.g., contracts and standards) and identifying non-compliance [1,5] (Sections 5–7 provide more details).

Some studies review AI applications in CS, which cover all AI aspects (e.g., computer vision and robotics) but lack comprehensive insights for NLP [1,6,7]. Besides, although NLP techniques are shared in different sectors where relevant review studies exist [8,9], the CS has unique data sources and applications. Thus, existing studies cannot summarise current achievements and provide advice for NLP-enabled smart construction. To bridge the gap, this paper provides a comprehensive review of main methods, applications, challenges, and trends of NLP in CS. In the paper, Section 2 introduces review strategies; Section 3 presents an overview of collected articles; Section 4 introduces bottom-level NLP techniques; Sections 5–7 present NLP-enabled applications; Section 8 discusses insights, challenges, and directions of NLP in smart construction; Section 9 concludes the study.

2. Data collection

The review strategy includes five steps. First, the scope is defined as NLP in CS. The second step determines databases and keywords for collecting articles. The Scopus and Web of Science (WoS) databases were selected due to wide coverage and well recognised quality. The query for searching the Scopus is shown below. A similar query with minor syntax modification was used to search WoS.

(construction PRE/5 project) OR (civil W/5 engineering) OR (civil PRE/5 project*) OR (construction W/5 manag*) OR (infrastructure*) OR (construction PRE/5 industry)) AND ((information W/5 extract*) OR (entit* PRE/2 recogni*) OR extract* OR mining OR reasoning OR retriev* OR (natural AND language AND process*) OR NLP) AND (text* OR document* OR "project specification*" OR reports OR records OR "contracts" OR "regulations" OR "files")*.

The query covers three aspects separated by '*AND*': 1) the scope (i.e., CS), 2) main NLP functions, and 3) common data sources. The query considers synonyms and word variations and combinations, e.g., manag* includes manage, manages, and management, and '*W/n*' specifies the number of words between two terms. The searching is restricted to journal articles to ensure quality. In the third step, 1288 articles were collected, and 492 duplicated ones were removed. The abstracts/keywords of the remained 796 articles were checked to ensure compliance with the scope, which left 124 articles for full-text review in the final step.

3. Overview of the results

3.1. Distribution of articles

Figs. 1-3 present article distributions based on years, journals, and regions, respectively. There is an overall increasing trend of NLP-related studies conducted for smart construction, which has become significant since 2015. This complies with other AI techniques (e.g., computer vision) in SC and is largely due to the increasing data needs for project management, enhanced capacity of computer hardware, and fast development and spread of AI techniques [6]. As for article sources, 54 journals published relevant articles. Most journals focus on the interdisciplinary sector of civil engineering, construction management, and computer science. Fig. 2 only shows journals with at least two publications. *Automation in Construction* is the most significant journal with 34 articles, about twice that of the journal ranked second (*the Journal of Computing in Civil Engineering*). Finally, the US plays a dominant role in this field, which has published one-third of reviewed articles and is followed by China, UK, and Korea. The remaining articles are distributed in Hong Kong, Taiwan, and several European countries (e.g., Spain and Italy).

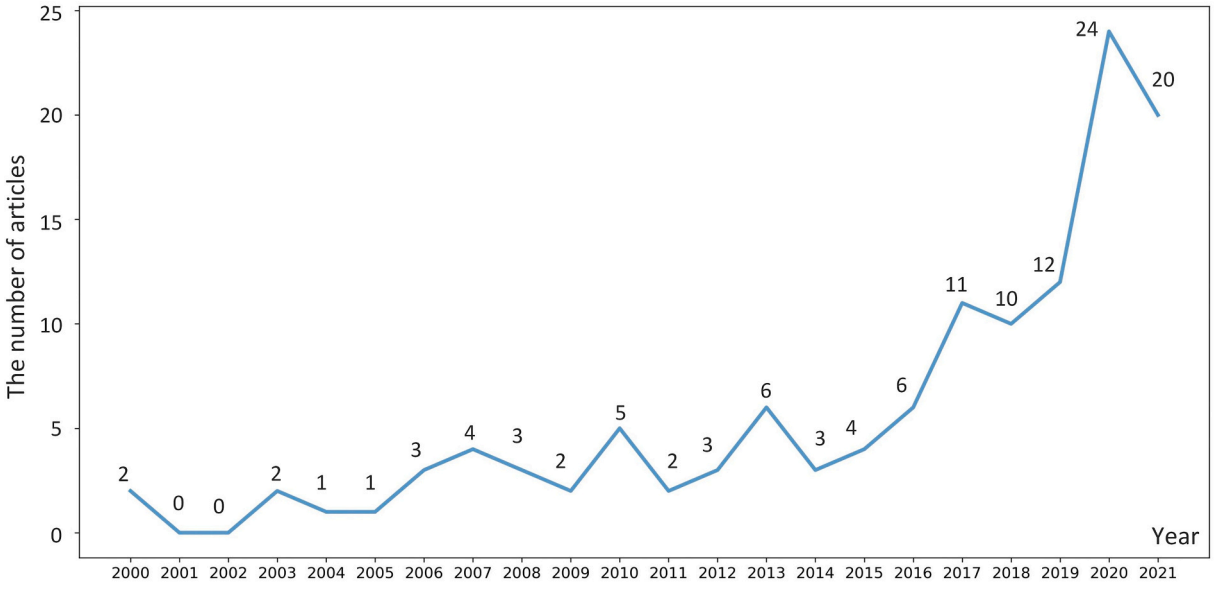


Fig. 1. Articles distributed by years.

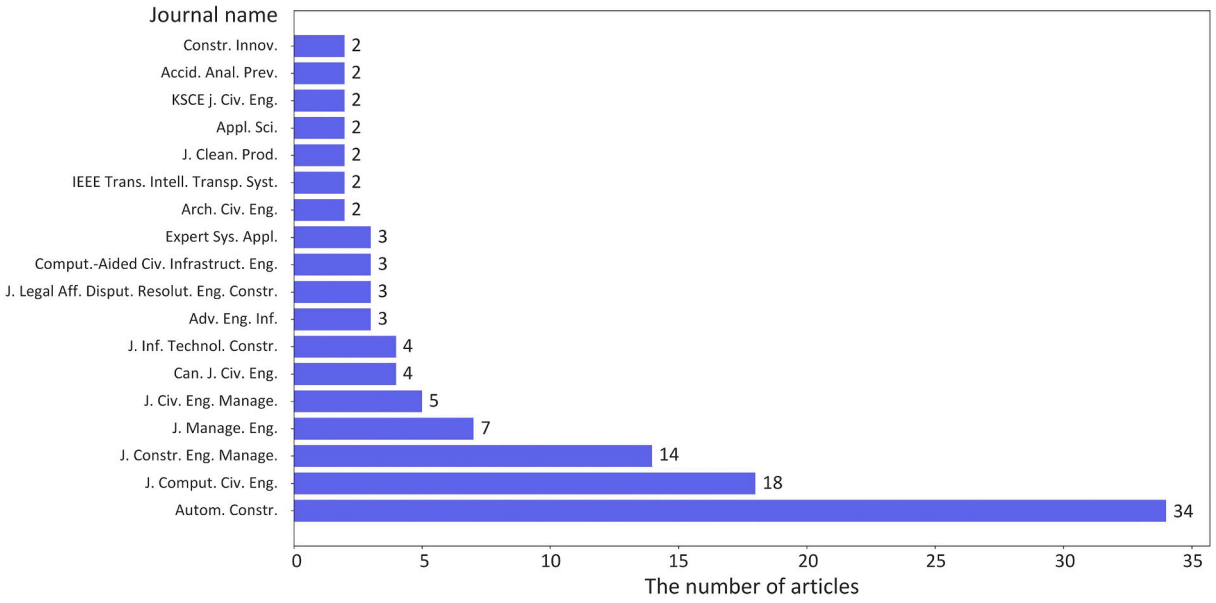


Fig. 2. Articles distributed by journals.

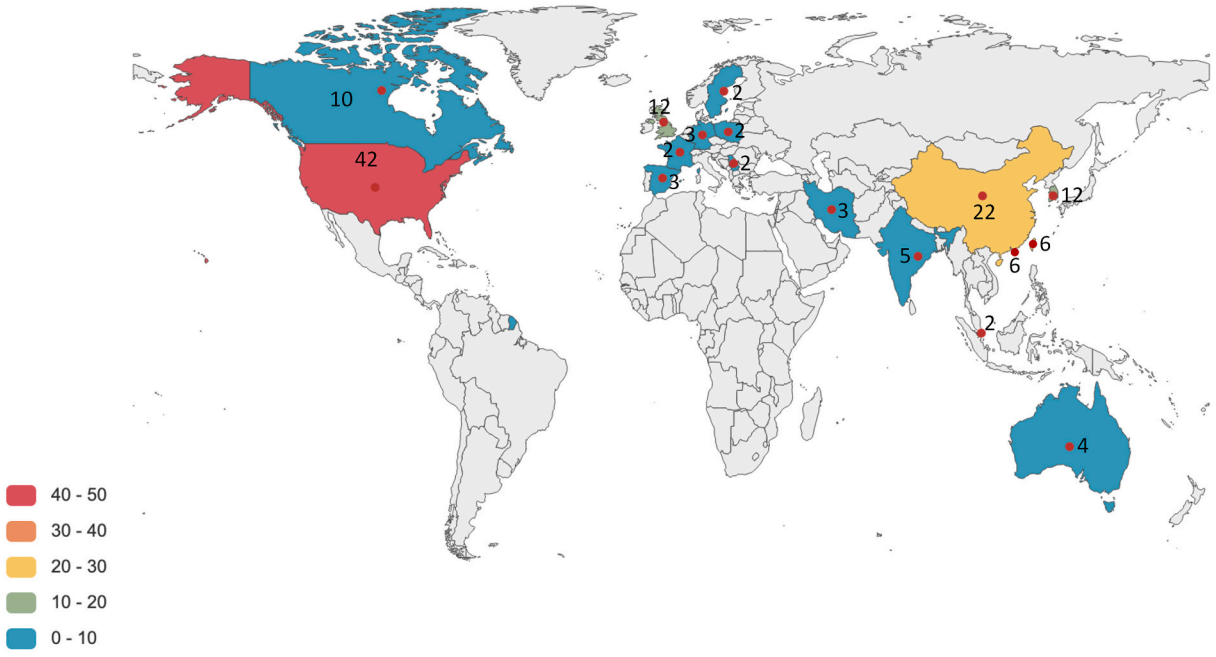


Fig. 3. Articles distributed by regions.

3.2. A knowledge map of topics

To gain an overall understanding of NLP in smart construction, abstracts of all reviewed articles are pre-processed and analysed with basic NLP techniques, i.e., frequency analysis of keywords and word cloud visualisation (Fig. 4). Details of the techniques are introduced in Section 4.

The results present distinct patterns. Keywords with the highest frequencies are 'information', 'text', and 'engineering documents', and 'management', indicating objects to which NLP is applied. They are followed by words of NLP techniques, e.g., 'mining', 'support' (representing support vector machines), and 'case' (representing case-based reasoning). Finally, words like 'extract', 'retrieval', and 'classification' indicate main NLP functions realised by the techniques. The word cloud also reveals keywords of certain applications, e.g., 'accident', 'risk', 'type', and 'semantic'. The structure of following sections (Fig. 5) is developed based on critical keywords identified in the exploration analysis. It consists of three main steps, i.e., NLP techniques and toolkits (Section 4), core functions that directly apply NLP to texts for extracting and exchanging important information (Sections 5–6), and downstream applications that rely on obtained information in the last step (Section 7).

4. Understanding NLP in construction projects

This section introduces the five core steps to apply NLP for smart construction: data collection, pre-processing, feature engineering, text analysing and understanding, and performance evaluation.

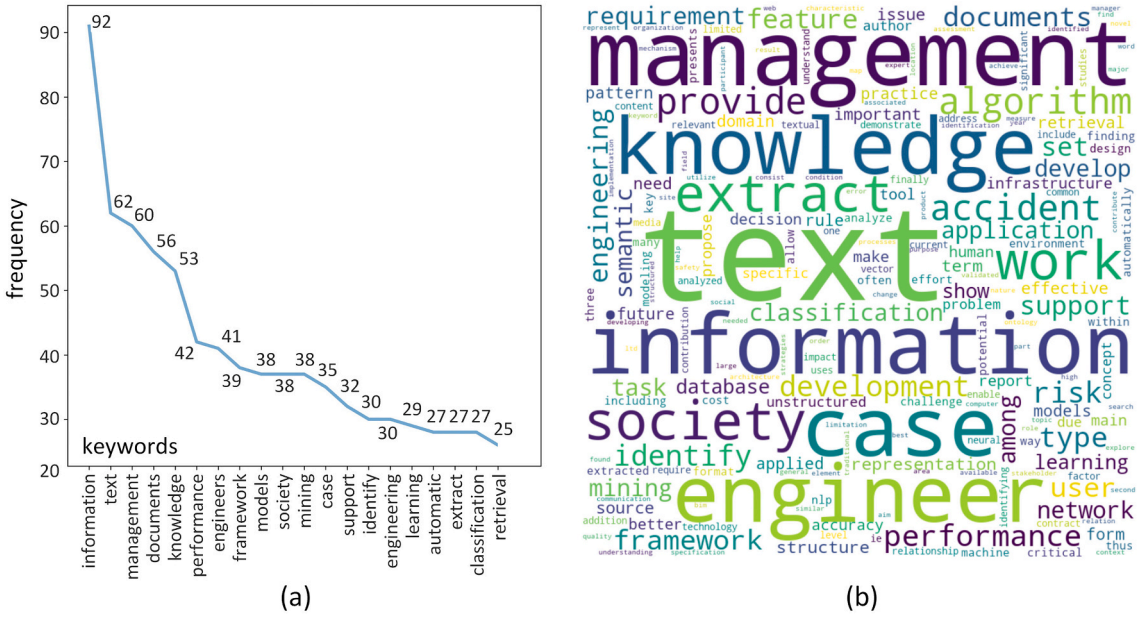


Fig. 4. Frequencies of keywords(a) and word cloud (b).

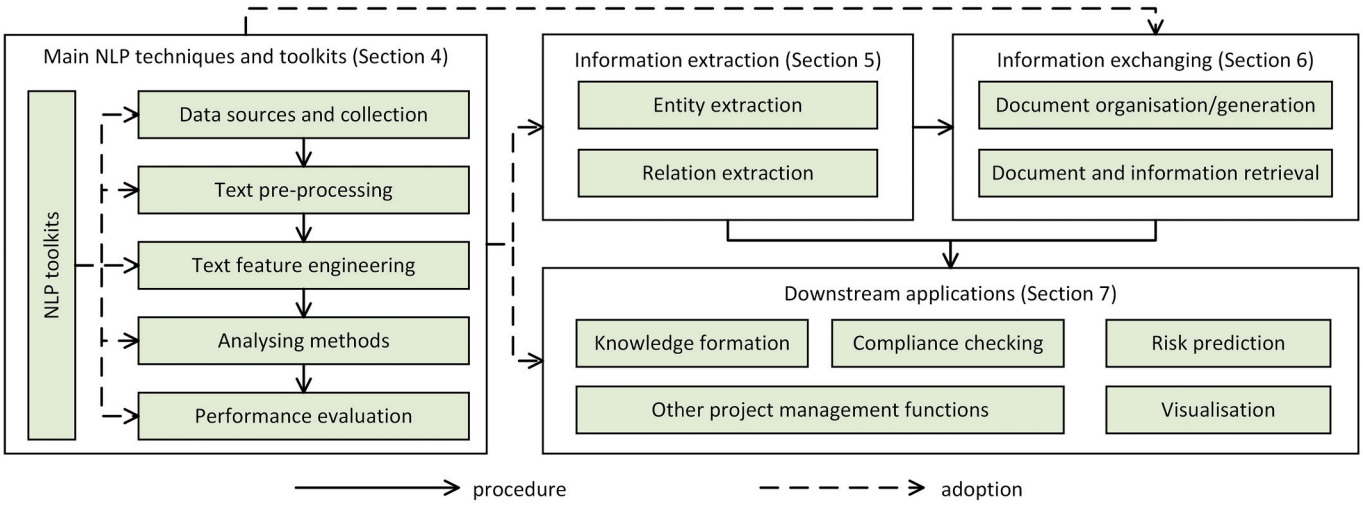


Fig. 5. Reviewed contents.

4.1. Data sources and collection

Data for NLP is mainly plain texts (called corpus). Table 1 lists dominant data types in CS. Data demands vary significantly in different articles, depending on data availability, problems faced and analysing methods. For instance, text data on social media is very large (e.g., millions of tweets) [10], while much less data (e.g., hundreds or thousands) is collected for project-level applications [11]. Besides, supervised ML/DL models require much more data than rule-based methods [12]. Most data is collected in three ways: 1) downloading from public databases, e.g., standards in government websites [13], 2) collecting from project teams [14], and 3) web-crawling [15]. All data is converted in digital forms, e.g., pdf, doc, and csv.

4.2. Text pre-processing

Freely written texts are very noisy. Hence, it is necessary to apply pre-processing for data cleaning. Table 2 introduces common pre-processing methods.

4.3. Text feature engineering

NLP methods rely on features hidden in texts. Moreover, computers can only handle numerical values. Thus, feature engineering is critical for NLP applications.

4.3.1. Text features

Text features consist of syntactic and semantic features. Syntactic features refer to arrangements and attributes of characters/words/phrases in sentences, e.g., a verb precedes a noun. Semantic features refer to semantic meanings of words/phrases, which can vary in different contexts. Common syntactic and semantic features are listed in Table 3.

4.3.2. Feature extraction and the usage of knowledge bases (KBs)

Most syntactic features can be directly extracted through open-source tools (Section 4.6). However, extracting semantic features needs domain knowledge, i.e., domain classes and their relations. Most studies define such knowledge in two types of KBs (Table 4).

Table 1
Data for applying NLP in smart construction projects.

Level	Source type	Examples
Project	Legislation documents	Contracts, bidding documents, claims, lawsuit case reports, dispute resolution records [16–19]
	Safety documents	Injury reports, incident narratives [12,20–22]
	Project management and evaluation documents	Cost/quantity records, working plans, drawings, site issue records, post-project reviews (PPRs), case reports, lessons learned records, disclosures [23–25]
	Maintenance documents	Maintenance records, inspection reports [26–28]
	Collaboration texts	*E*-mails, meeting agendas, requests for information (RFIs), chatting records, design change records [29–31]
External	Online resources	News, advertisements, web-pages, patents, research articles [32–36]
	Social media texts	Twitter, WeChat [36–38]
	Regulatory documents	Standards, manuals, specifications, guidelines, codes [39–42]

Table 2
Summary of text pre-processing methods.

Type	Method	Description
Separation	Sentence splitting	Identify individual sentences based on boundaries, e.g., periods.
	Tokenisation	Divide a sentence into separated tokens. A token can be single words/characters or phrases.
Cleaning	Normalisation	Remove non-textual contents (e.g., images) and non-relevant texts, e.g., texts of project team are removed for safety analysis.
	Length-based removal	Remove sentences/words/characters shorter than a certain length.
	Frequency-based removal	Remove words/characters with frequencies less than a threshold.
	Punctuations/ stopwords removal	Remove punctuations and words/ characters without significant meanings (e.g., 'and', 'the'), which is often based on published dictionaries recording common stopwords.
Modification	Lowercasing	Change all words/characters to lowercase.
	Stemming or lemmatisation	Convert words to standard forms. Stemming directly chops off ends of words; lemmatization returns words to dictionary forms using a vocabulary or morphological analysis.
	Sentence simplification	Convert complex sentences (e.g., nested or parallel forms) to simple sentences using heuristic rules.
	Extraction	Extract texts from particular formats (e.g., XML or HTML), which is often applied to online data.

Extracting semantic features with KBs requires mapping between tokens/n-grams and KB classes based on existence checking or cosine similarity [58,67]. Once the mapping is finished, apart from the mapped class, additional semantic features (e.g., synonyms and negations) could be inferred to enrich rules and vectors (Section 4.4) by navigating KB relations, e.g., generalisation/specialisation, composition/aggregation, logical relations (e.g., disjoint or equivalent), and concept association (e.g., a beam supports a floor) [60]. The semantic features can facilitate method development and other applications. More details are presented in Section 4.4 and Sections 5–7, respectively. Fig. 6 shows the mapping with a simple example, i.e., mapping the entity 'tower crane' to the class 'Crane' while inferring its superclass 'Equipment' and related class 'Lifting Task' as semantic features.

However, manually building KBs is labour intensive and time-consuming, which fortunately can be to-some-extent automated using

Table 3
Summary of text features.

Type	Feature	Description and examples
Syntactic	n-grams	The number of words/characters in a token, e.g., unigram, bigram, and trigram indicate tokenisation returns tokens containing one, two, and three words/characters, respectively [29,43,44]
	Frequencies	The frequency of n-gram tokens [34,45,46]
	Part-of-speech (POS)	Word types (e.g., verbs, nouns) distinguished based on ideas denoted and functions performed in a sentence [42,47].
	Phrase-grammatical structure (PSG)	Labels of short phrases of tokens/n-grams (e.g., noun phrases, verb phrases) [40,48]
	Dependency grammar	Governor-dependent relationships among tokens/n-grams, e.g., 'amod' (actual, condition) indicates 'condition' is governed by the adjective word 'actual' [47–49]
Semantic	Domain classes	Concepts of tokens/n-grams in knowledge bases (KBs) (e.g., the class of 'tower crane is 'Crane'). KBs are elaborated in the next section [50,51]
	Synonyms/antonyms	Words having similar (e.g., 'subway' and 'metro') and opposite meanings (e.g., 'success' and 'failure') [52–54]
	Negations	Words with negative connotation (e.g., no, not) [39,55]
	Hierarchical or semantic relations	A tree structure among KB classes (e.g., super- and sub-classes) or network structure when semantic-rich relations (see Section 5.2 for more details) are added [53,56]

Table 4
Two main KB types.

Type	Explanation	Examples
KBs without relations	Group of distinct domain characters/words/phrases	Thesaurus, lexicons, dictionaries, vocabularies, gazetteers, and lists [57–61]
KBs with relations	Interconnected domain classes	Meta-models (e.g., International Framework for Dictionaries (IFD) and Industry Foundation Class (IFC)) [62–64], taxonomies and hierarchies [56], ontologies [50,51,59,60], and knowledge graphs (KGs) [26,65,66]

NLP. To build KBs without relations, key entities can be extracted (see Section 5 for more details) and directly modelled as classes. However, auto-developing KBs with semantic-rich relations (e.g., ontologies) is challenging [68]. Many studies are limited to modelling taxonomical relations (i.e., sub-class) by using hierarchical clustering to group entities [59]. For instance, Chi et al. [56] proposed an iterative expansion approach to create taxonomies for seismic engineering. The top-level classes were used as queries to retrieve documents, whereas lower-level classes were derived from top-terms in retrieved documents based on the terms' correlation (Chi-square) with higher-level classes (query terms).

4.4. Text analysis and understanding

Text features extracted in Section 4.3 can already enable simple NLP applications (e.g., extracting tokens/n-grams as keywords if their frequencies exceed a threshold). Nevertheless, sophisticated text analysis requires more complex methods which are introduced in this section. Besides, additional methods worth introducing are additionally discussed in Section 5–7.

4.4.1. Heuristic rule-based methods

Heuristic rules are widely applied to extract entities/relations, exchange information, and compliance checking by matching pre-defined

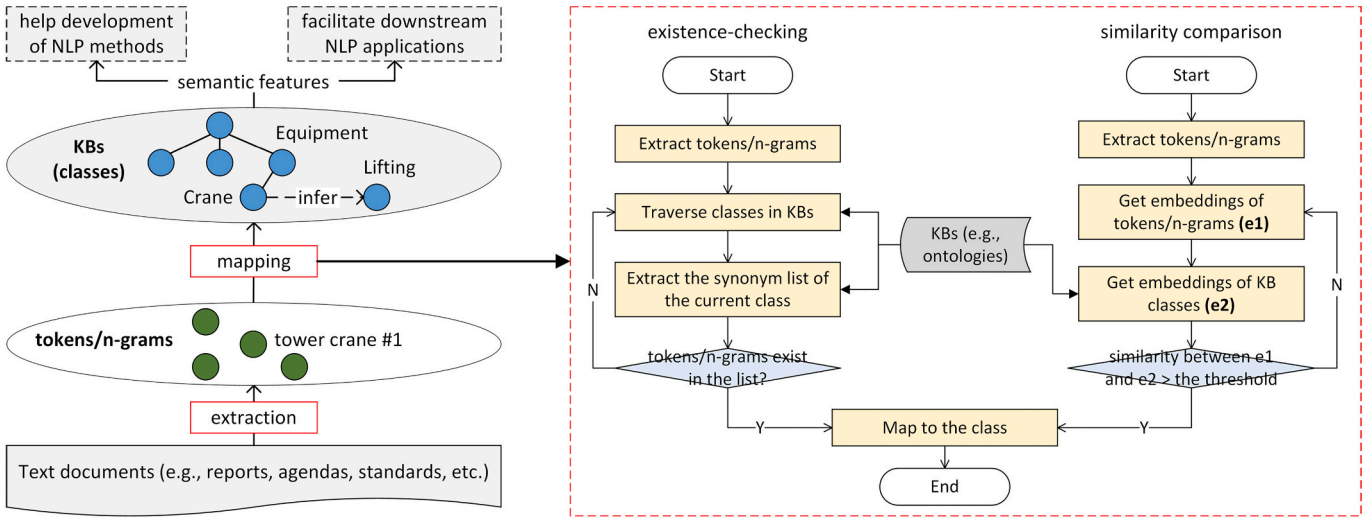


Fig. 6. Mapping between KBs and tokens/n-grams.

patterns. Rule development has three levels.

Simple rules are developed on basic features (e.g., POS/PSG tags and orders of tokens/n-grams) [49,69,70] and tuple-patterns formed by several tokens/n-grams (e.g., subject-verb-object (SVO) tuples) [13,71]. Tuple-patterns are mainly employed for extracting relations (Section 5.2). For example, a simple rule for extracting entities in contracts is 'if a noun is followed by a modal word and a basic verb, then the sentence is a clause, and the noun is the clause subject'. However, simple rules suffer from low expressiveness thus can be improved by adding features listed in Table 5.

Simple rules can be combined as complex rules by logic operators (e.g., AND, NOT, OR). The logical components, together with semantic features, enable reasoning to evaluate applicable conditions and exceptions of rules [76,81–83]. For example, when querying provisions (Section 6.2) of the 'lifting' action, a complex rule can deduce that 'crane' (implicit) is associated with explicit entities 'lifting' and 'bucket' in the query. The query therefore is expanded as provisions about 'crane' are evaluated as applicable [84]. Finally, special rules (e.g., conflict resolution rules) are needed to handle exceptions. For instance, entity extraction can be confused when two words representing comparative relations (e.g., above and not less than) are matched; in this case, priority rules (e.g., a higher priority is set for 'not-less-than') are applied to solve the conflicts and extract the entity following 'not-less-than' [58,69]. Heuristic rules can reach high precision. However, patterns for rule development are restricted to certain corpus, thus, rules often suffer low recall and poor generalisation ability [7].

4.4.2. *Vectorisation*

Many analysing methods, from simple cosine similarity computing to complex ML/DL models, can only take numerical vectors as inputs. Thus, this subsection introduces two main approaches that can convert texts to vectors as inputs for text analysing. The first approach is frequency-based, which constructs a vocabulary to cover all $\boldsymbol{V}$ unique tokens/n-grams and their frequencies in the corpus containing $\boldsymbol{N}$ documents. As shown in Fig. 7, each document and the corpus are converted into a vector $\in\mathbb{R}^{V}$ (i.e., bag-of-words (BOW) or term-frequency (TF) vector), and a matrix $\in\mathbb{R}^{D\times V}$ (i.e., term-frequency (TD) matrix), respectively [85]. The vectors can be enhanced by weighting and enriching. For instance, Eq. (1) assigns more weights to tokens/n-grams with high frequencies in individual documents and low frequencies in the entire corpus, where tf_d and tf_d represent term-frequencies in one document and the corpus, respectively. This can recognise distinct tokens/n-grams and produce the widely-adopted term-frequency-inverse-document-frequency (TF-IDF) vectors [86]. More sophisticated weighting is applicable to vector elements based on their length/position (e.g., if a token/n-gram is in section heads) [28,65,87,88], information entropy, mutual information, and correlation (e.g., Chi-square) [35,41,53,56]. Meanwhile, vectors enriching adopts similar features of rule enhancement, which appends syntactic/semantic features (e.g., the KB class and number of grammatical dependencies of a token/n-gram) to the original vector [54,62,64,89,90].

$$TF-IDF = log(tf_d+1)\times\left[1+log\left(\frac{N}{tf_N}\right)\right] \quad (1)$$

The second approach (*word2vec*) generates text vectors by capturing their semantics (called word embeddings). As illustrated in Fig. 8, this is realised by ML models trained with 'contexts' (i.e., some tokens/n-grams surrounding a central token/n-gram. Two training strategies (i.e., CBOW and skip-gram) are applicable. CBOW takes context tokens/n-grams as inputs and predicts the central token/n-gram, while skip-gram reverses the roles [8,53]. Besides, the *doc2vec* approach has been proposed to generate embeddings for longer texts (e.g., paragraphs) [91]. However, training is data-demanding and time-consuming. Thus, many studies use published embeddings pre-trained on large databases (e.g., Wikipedia) and fine-tune the values in downstream applications [44,92].

Finally, the obtained vectors/matrices are usually large and sparse. Hence, many studies implement decomposition methods to identify data eigenvectors and reduce dimensions, including singular value decomposition (SVD), principal component analysis (PCA), latent semantic index (LSI), and latent semantic analysis (LSA) [17,85,90,93]. It has been proved these techniques can increase efficiency and accuracy in subsequent analysis [94].

Table 5
Features for rule improvement.

Feature type	Explanation/examples
Syntactic features	Regular expressions [58,72,73] Distance between tokens/n-grams Positions of tokens/n-grams [24,74] Special labels of online texts (e.g., XML/HTML tags and URLs) [34,75]
Semantic features	Classes, relations, and synonyms defined in KBs [66,76,77] 'Trigger' words, such as negations, modal words (e.g., shall, must), and transition words (e.g., because, but), which are useful for regulatory documents and contracts [78–80]
Latent features	Weights according to expert opinions [24,25,34,66,75,77]

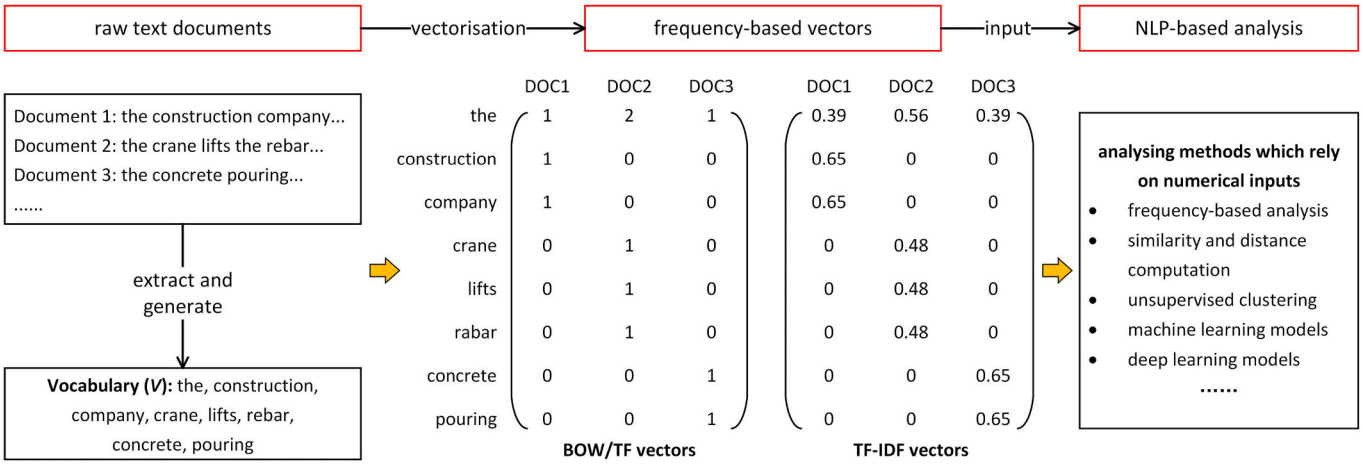


Fig. 7. Frequency-based vectorisation [3].

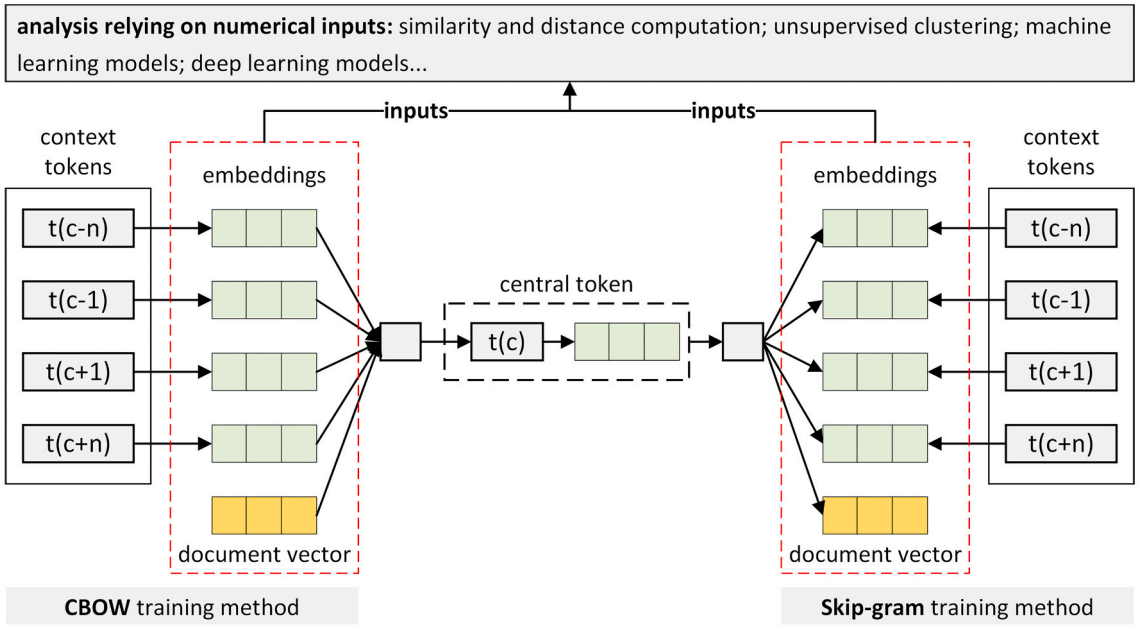


Fig. 8. ML-based vectorisation [8].

4.4.3. *Unsupervised methods*

- **Distance computation and clustering**

Computing the distance between two entities is important for NLP-enabled applications in CS (e.g., finding the most relevant documents). Common distance measures include Euclidean distance (Eq. (2)), cosine distance (Eq. (3)), Manhattan distance (Eq. (4)), and Chebyshev distance (Eq. (5)), where m is the number of vector elements [95]. Distance is also the basis of clustering which is an effective method to partition data into clusters based on the distance to cluster centroids (Fig. 9(a)). Clustering is mostly used for organising documents (Section 6) and discovering knowledge (Section 7.1).

$$D(u,v) = \sqrt{\sum_{i=1}^{m}(u_i - v_i)^2} \tag{2}$$

$$D(u,v) = 1 - u \bullet v / |u||v| \tag{3}$$

$$D(u,v) = \sum_{i=1}^{m}|u_i - v_i| \tag{4}$$

$$D(u,v) = \underset{i}{\operatorname{argmax}}(|u_i - v_i|.) \tag{5}$$

- **Topic modelling**

Topic models take raw documents and/or their topics as inputs then identify themes from texts by mimicking document generation, which is very useful for knowledge formation (Section 7.1), e.g., deriving root causes of accidents from injury reports. Latent Dirichlet Allocation (LDA) is the most popular topic model in CS. As shown in Fig. 10, LDA works by deriving four distributions based on text statistics, i.e., two Dirichlet distributions for topic-document association and two multinominal distributions for words-topic association. Variants of LDA, e.g., Dynamic Topic Model (DTM) and Social Topic Model (STM), considering correlation and dynamics of words and documents, are also proposed to improve modelling results [32].

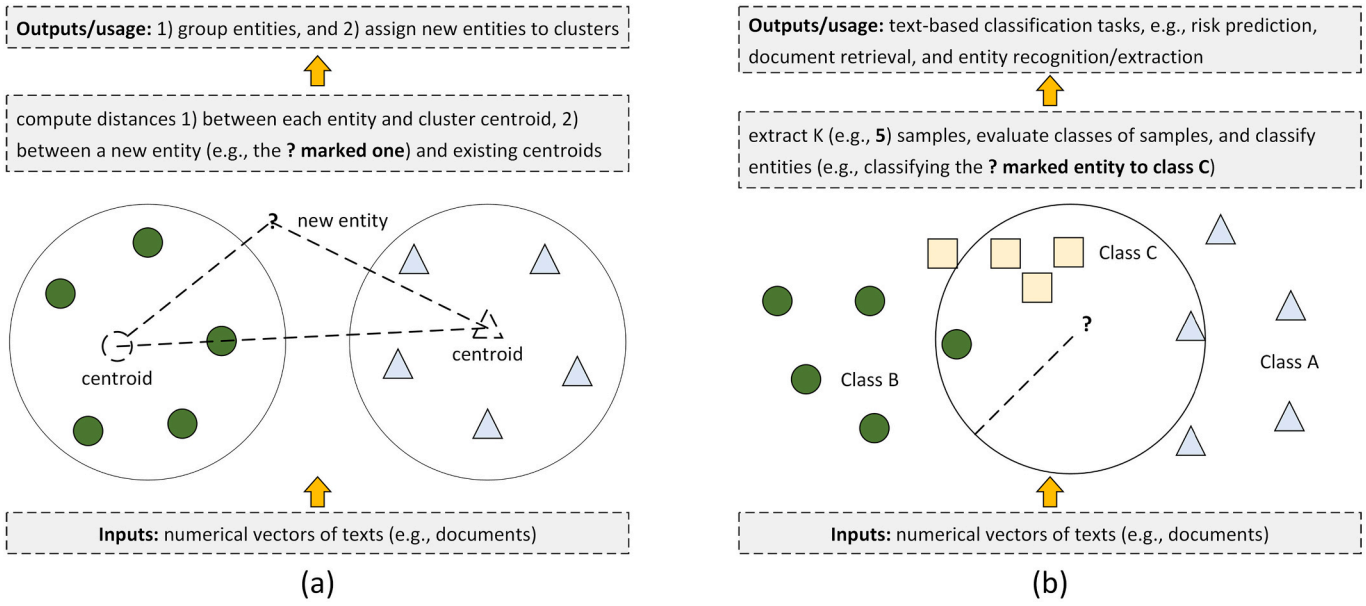


Fig. 9. Distance-based methods (a) clustering, (b) KNN (taking document grouping/classification as an example) [4]

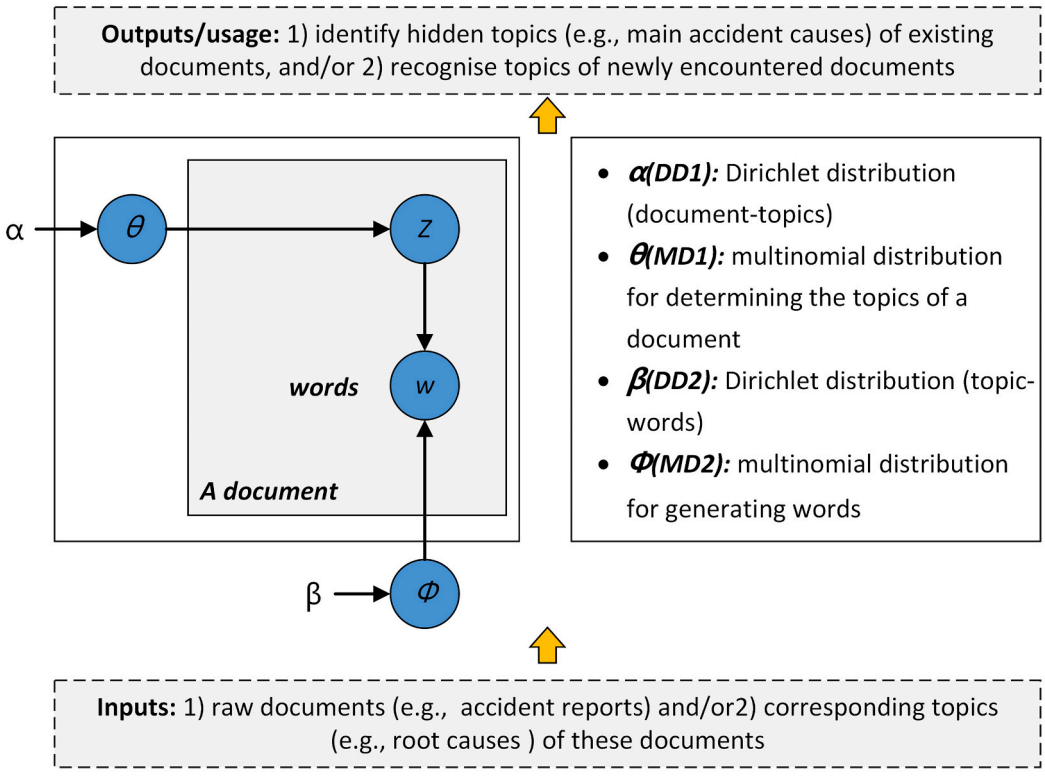


Fig. 10. LDA modelling (taking accident causes identification as an example) [4]

- **Association rule mining**

Association rule mining discovers data dependency patterns based on three measures (confidence, support, and lift). The measures are computed through Eqs. (6)–(8) and reflect the probability of patterns $A \rightarrow B$ appearing in texts. Apriori, which tests candidate patterns with grid searching and forms rules if their measures exceed certain thresholds, is the main rule mining algorithm [96].

$$Confidence(A \rightarrow B) = P(A \cup B)_{/P(A)} \quad (6)$$

$$Support(A \rightarrow B) = P(A \cup B) \quad (7)$$

$$Lift(A \rightarrow B) = P(B|A)_{/P(B)} \quad (8)$$

4.4.4. Supervised ML/DL models

Unsupervised methods are easy to implement while often suffer low accuracy and instability. Thus, many NLP studies employ supervised models. Most supervised models are applied for classification tasks, e.g., identifying the types of entities, relations, and documents for

information extraction and exchanging (Sections 5–6), classifying construction tasks and resources, and recognising affecting factors and types and severity of risks and damages (Section 7).

- **Naïve Bayes (NB)**

Naïve Bayes aims to find joint probabilities of data features and classes which are assumed to be independent. The model is trained by estimating statistics (Eq. (9)), where d_i and $\boldsymbol{C}$ indicate the ith feature and class of a sample, respectively. Then, new data is assigned to classes $\boldsymbol{C}$ with the highest probability, i.e., $argmax(P(\boldsymbol{C}|d))$ [97].

$$P(\boldsymbol{C}|d_i) = P(d_i|\boldsymbol{C}) \times P(\boldsymbol{C})_{/\boldsymbol{C}} \; i = 1, 2\ldots \tag{9}$$

- **Decision tree (DT) and random forest (RF)**

DT is a classification algorithm which builds a tree by splitting a feature (i.e., a node) into sub-nodes (Fig. 11). According to Eqs. (10)–(12), DT splits the feature which results in most information gain ($\boldsymbol{G(S)}$), where $\boldsymbol{D}$ and $\boldsymbol{D_j}$ are the dataset and subset ($\boldsymbol{S}$) split on j^{th} feature (suppose there are k features and m samples) [98].

$$H(X) = -\sum_{i=1}^{m} p_i \times log_2(p_i) \tag{10}$$

$$H_S(D) = -\sum_{j=1}^{k} \frac{|D_j|}{|D|} \times H(D_j) \tag{11}$$

$$G(S) = H(D) - H_S(D) \tag{12}$$

- **K-near neighbour (KNN)**

Similar to clustering, KNN classifies data based on data distance in the training set (Fig. 9(b)). During testing, KNN: 1) finds $\boldsymbol{K}$ samples in the training set with the closest distance to the new data, 2) assigns new data to the class containing most samples among the $\boldsymbol{K}$ samples [30].

- **Support vector machine (SVM)**

Some data cannot be distinguished in low-dimensional spaces. SVM adopts kernel functions f to project data into a high-dimensional space. SVM finds a hyperplane that 1) separates data according to the distance $\boldsymbol{d}$ to the hyperplane and 2) maximises margin γ between classes by solving the optimisation problem (Fig. 12 and Eqs. (13)–(16)) [4]. Due to the unique projection mechanism, SVM can gain better classification performance than other ML models and even DL models, which is the most popular classifier in NLP-based applications (more details are provided in Sections 5–7).

$$\gamma = 2/\|w\| \tag{13}$$

$$d = f\left(w^T x + b\right) \tag{14}$$

$$minimise_{w,b} = 1/2\|w\|^2 \tag{15}$$

$$subject\ to = y_i\left(w^T x_i + b\right) \geq 1, for\ i = 1, 2\ldots \tag{16}$$

- **Ensemble model**

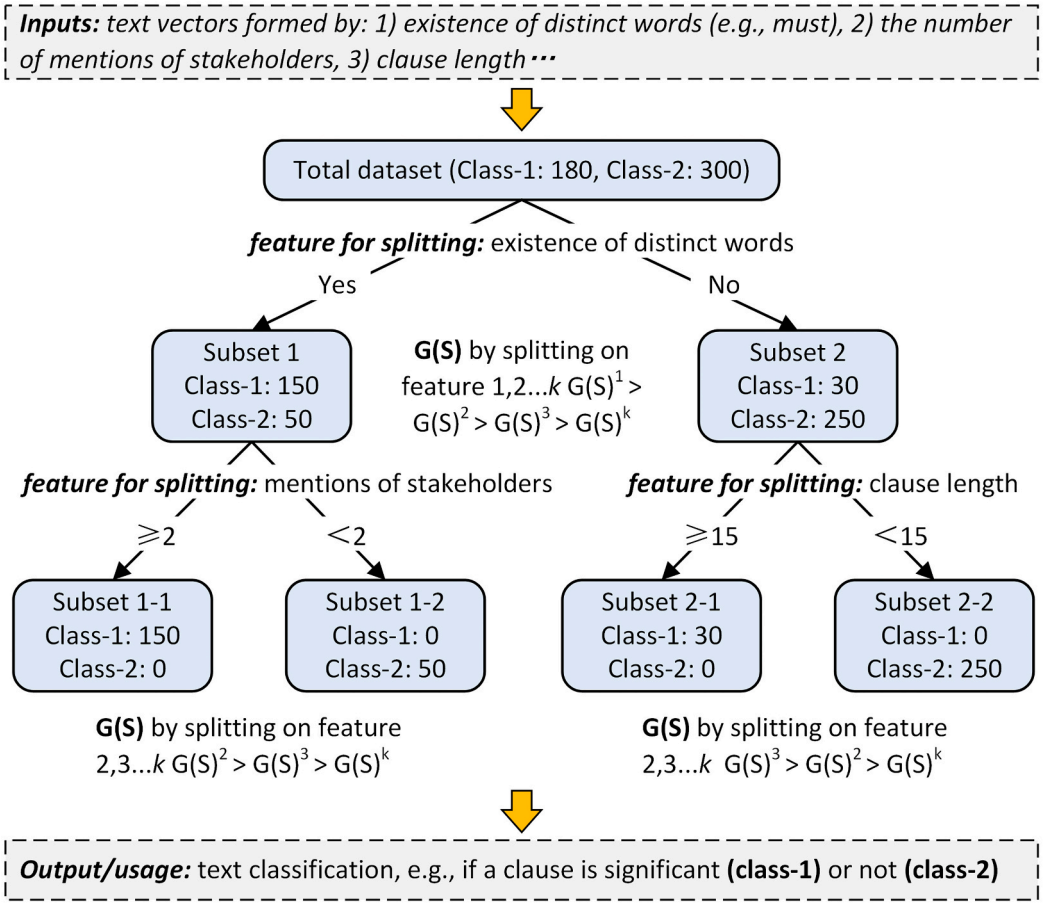


Fig. 11. DT mechanism (taking contract clause classification as an example) [98]

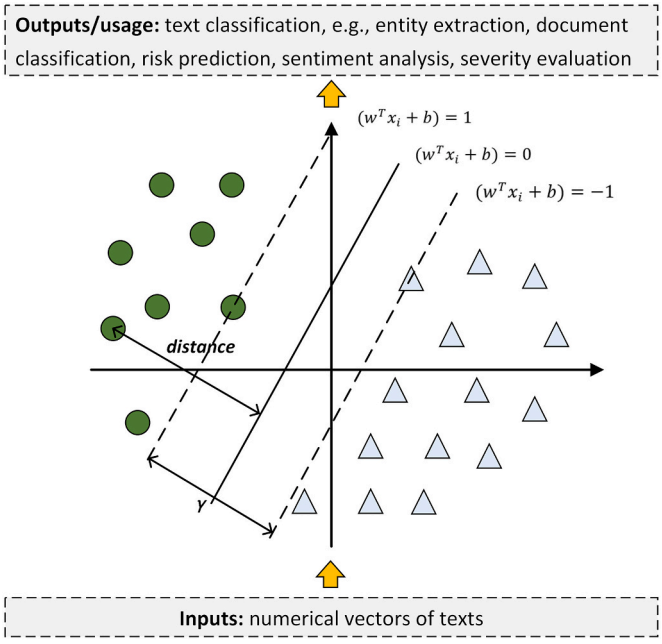


Fig. 12. SVM mechanisms [99]

ML models have unique advantages/disadvantages. Ensemble models can combine individual models to form a stronger classifier. In NLP studies, popular ensemble models include Bagging, AdaBoost, gradient boosting and regression tree (GBRT), and Random Forest (RF) for DT. The final predictions are mainly made by voting, averaging, and quadratic programming [41,98].

- **DL models**

The main drawback of ML models is the reliance on manual feature engineering. This is very time-consuming and hinders model generalisation as features are restricted to certain datasets. Therefore, recent studies in CS train DL models to extract features automatically. The three widely applied DL-NLP models are the gated recurrent unit (GRU) [43], long-short-term-memory (LSTM) [68], and convolutional neural network (CNN) [92,99]. The difference is that CNN uses a sliding window to scan texts and extract features with convolution and pooling operation, while GRU/LSTM process texts sequentially, which thus can better memorise information in early steps; Fig. 13 shows the working mechanisms, and interested readers can refer to [100]. In NLP-based applications, CNN is more applied to long texts (e.g., sentences); LSTM/GRU focuses on individual words/characters and is especially useful for entity extraction; and both are used for risk prediction and relation extraction (Sections 7.3–7.4).

4.4.5. *Case-based reasoning (CBR)*

CBR is a data-driven information searching method, which retrieves the most similar case based on the query. Thus, CBR is usually applied to retrieve documents given searching terms (Section 6.2) or finding questions or solutions given the question (Section 7.3). CBR involves four steps (Fig. 14) and depends on the distance between query-case vectors. The vectors can be formed with text features (e.g., BOW) [54], categorial attributes (e.g., if a risk factor exists), and combined information [101].

4.4.6. *Network analysis*

Network is another useful NLP-related tool in smart construction. Networks are created by either co-occurrence relations (e.g., social network) or semantic-rich relations (e.g., material constrains task), which can derive distinct patterns (e.g., causality and dependencies) thus help knowledge formation (Section 7.1) and project decision making (Section 7.4). Section 5.2 introduces more details of these relation types. Nevertheless, network analysis relies on certain measures

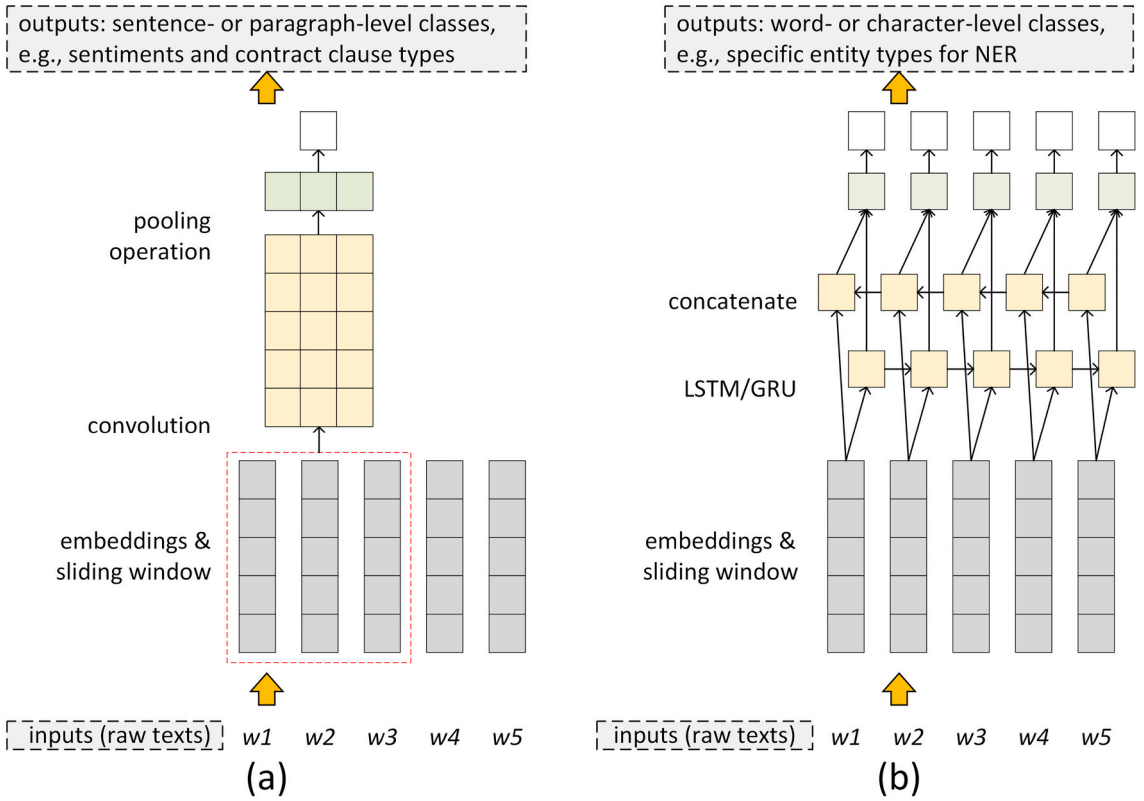


Fig. 13. Mechanisms of DL models (a) CNN, (b) GRU/LSTM [100]

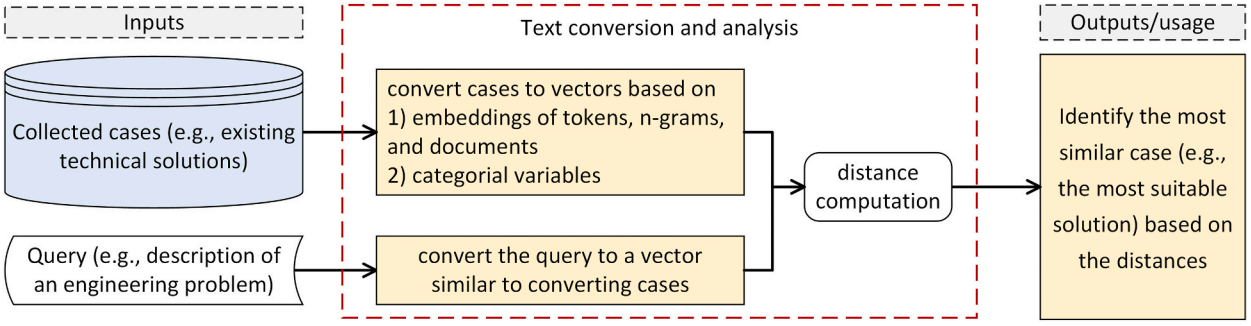


Fig. 14. CBR mechanism (taking technical solution retrieval as an example).

and is conducted at three levels (Table 6).

4.5. Performance evaluation

Above analysing techniques can only be applied when appropriate performance evaluation methods are in place, and Table 7 lists main quantitative performance measures.

4.6. NLP toolkits

There are many tools realising NLP functions. As shown in Fig. 15, some tools are developed for specific functions, whereas others encapsulate multiple functions. Besides, some tools/packages are developed using specific languages, and others can be implemented by multiple languages.

Based on the review of NLP techniques, two main ways are identified for increasing performance (especially accuracy), i.e., applying more sophisticated ML/DL models and integrating more domain information or knowledge as additional features.

5. Information extraction

Freely written text data is usually unstructured and noisy. Therefore, useful information must be identified and extracted before being exchanged (Section 6) or fed into downstream applications (Section 7). Such information commonly takes the form of entities and relations among entities. As mentioned at the end of Section 3 and illustrated in Fig. 5, this section introduces one core NLP function, i.e., extracting semantic-rich entities and relations from texts.

5.1. Entity extraction

5.1.1. Entity types

Entities are tokens/n-grams carrying important meanings. As revealed in Fig. 16, the procedure identifies useful entities and discards other tokens/n-grams [95,107]. Table 8 presents main entity types and their sources.

Table 6
Network measures and usages.

Level	Common measures	Usages
Node	betweenness, centrality, in−/out-degree, PageRank, and follower/following ratio for social media [37]	Identify critical nodes, classify nodes, predict missing nodes
Edge	average path length and path exploration [65]	Identify mutual impact and causal relations, predict missing links
Graph	centralisation, density, and topology (i.e., node connections) [102]	Cross-compare, partition, and classify graphs

Table 7
Main performance measures.

Task type	Measures
Classification	Label measures: true positives (TP), false positives (FP), true negatives (TN), false negatives (FN), Precision (P) (i.e., sensitivity), Recall (R), Specifity (S), Accuracy (ACC), and F1-score (F1), Kappa value, and Area Under Curve (AUC) [25,26,40,86] $P = \frac{TP}{TP+FP}$, $R = \frac{TP}{TP+FN}$, $S = \frac{TN}{TN+FP}$, $F1 = \frac{2 \times P \times R}{P+R}$ For multi-label classification, macro-P, macro-R, and macro-F1 should be used, where P, R, F1 are computed for each label and averaged $ACC = \frac{TP+TN}{TP+TN+FP+FN}$ $Kappa = \frac{ACC-pe}{1-pe}$ where $pe = \frac{\sum_{i=1}^{n} sum(row(i)) \times sum(col(i))}{\forall i \in C\ (sum(i))^2}$ AUC: area under the Receiver Operating Characteristic (ROC) curve formed by different P and R values
Information extraction/retrieval	Label measures: P, R, F1, similar to classification measures [90,103,104] Ranking: true relevant items in the list of all items [46] Efficiency: time, and consumed memory for retrieval [37]
Regression	Difference error measures: mean square error (MSE), root mean square error (RMES), sum of square regression (SSR), sum of square total (SST) [29,101,105] $(MSE) = \frac{1}{N}\sum_{i=1}^{N} w_i (y_i - \hat{y}_i)^2$, $RMSE = \sqrt{MSE}$, $SSR = \sum_{i=1}^{N} w_i (\hat{y}_i - \bar{y})$ $SST = \sum_{i=1}^{N} w_i (y_i - \bar{y})$, $R^2 = \frac{SSR}{SST}$, $\hat{y}_i$, y_i, and $\bar{y}$ are predicted, true, and average of true values, respectively
Clustering	Distance measures: clustering is more coherence when data have smaller and lager distance when they are in the same and different clusters, respectively. The widely used measures include Jaccard index, KL divergence silhouette coefficient and DB index [28,32,106] Entropy measures: Perplexity (mainly used for topic modelling) [22,32] Label measures: P, R, F1, when labelled clustering results are available [37,96]

5.1.2. Entity extraction approaches

Based on methods introduced in Section 4, Table 9 summarises four ways to extract entities.

5.2. Relation extraction

5.2.1. Relation types

As shown in Fig. 16, relation extraction (except relation names) includes two levels: tuples and semantic relations. Tuple extraction groups tokens/n-grams into chunks, only implying existence of relations among tokens/n-grams. Semantic relation extraction recognises explicit and semantic-rich relations among two or three tokens/n-grams. Table 10 reveals main relation types to be extracted.

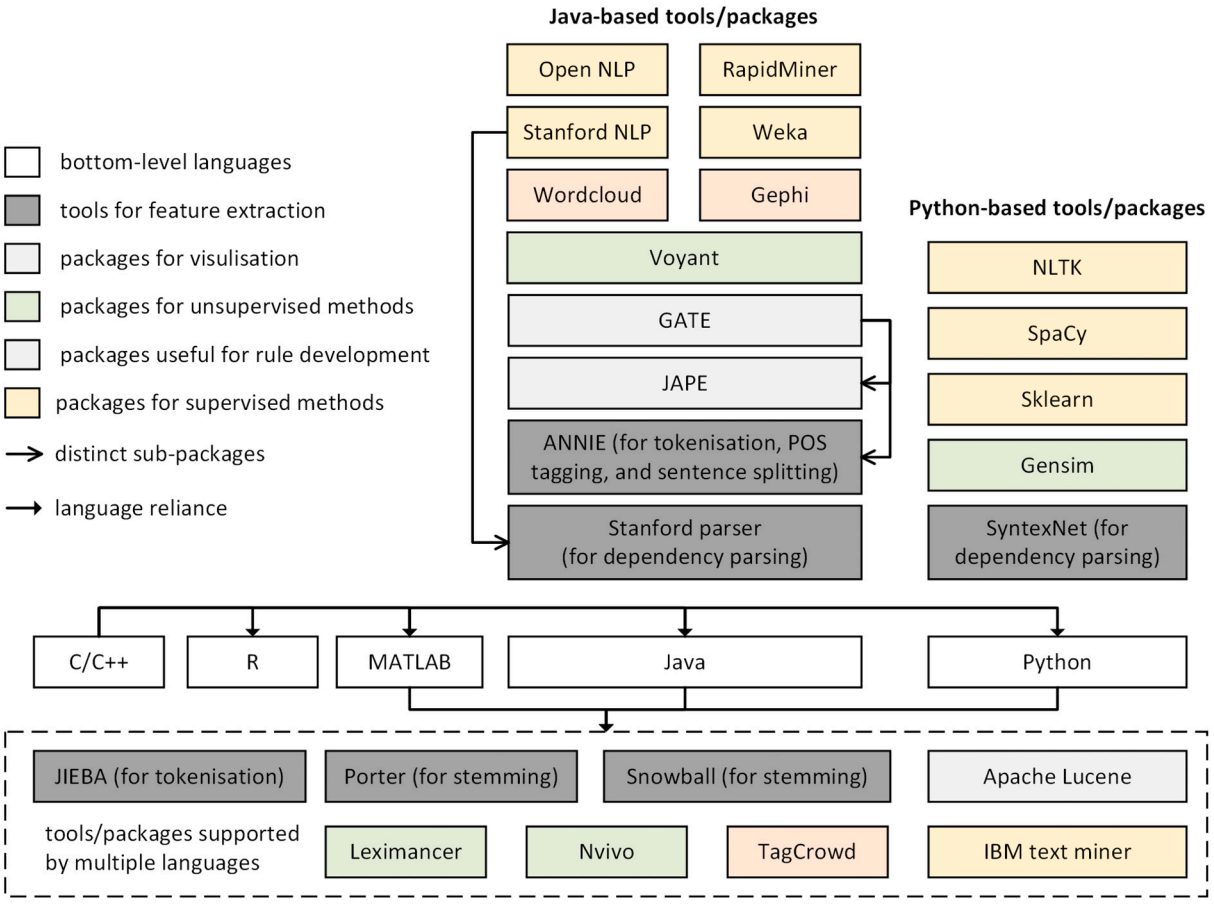


Fig. 15. Mainstream NLP toolkits.

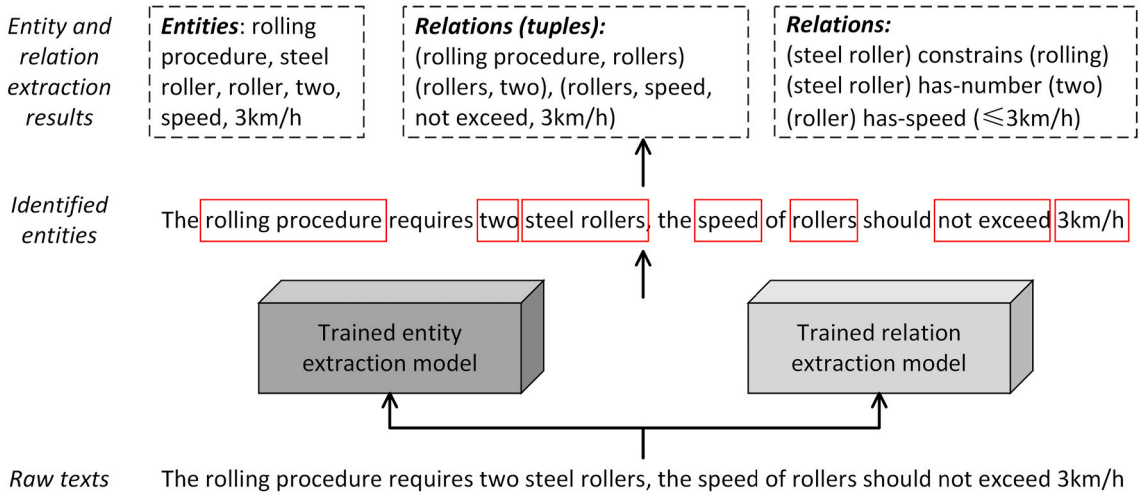


Fig. 16. Conceptual mechanism of entity and relation extraction.

5.2.2. Relation extraction approaches

Table 11 presents major relation extraction methods. It is worth noting that as an early attempt for extracting semantic-rich relations in CS, Wu et al. [68] trained knowledge representation learning (KRL) models to extract constraining relations among project entities and reached an accuracy over 80%. KRL is a cutting-edge technique for extracting relations of triple forms (head-relation-tail or h, r, t) by assuming $t \approx h + r$ for valid training triples, where word embeddings of entity/relation names are adopted to represent h, r, t while triple features are extracted by CNN. Interested readers can refer to [119].

Despite of the achievements of NLP-based information extraction, they focus on entity extraction. Research for relation extraction is inadequate, and only relations with basic semantics (e.g., similar-to and hypernym) can be extracted [2].

6. Information exchanging

The information extracted from texts using methods introduced in

Table 8
Entity types.

Entity types	Examples
Safety accident/incident information	Accident types, root causes, injured body parts, outcomes [35,58,70,105]
Safety hazards	Hazardous objects, locations [73,81]
Common risk factors	Change order, policy, time (e.g., season and day), weather, site condition, and market [18,87]
Obligations and requirements	Shall not subcontract the whole work, shall comply with engineers' instruction [76,78,108]
Project stakeholders	Contractor, supplier, and owner [78,109]
Project products and facility information	Beam, column, and floor, landmark, facility equipment state, location and type [14,27,49,57,88,110,111]
Project execution information	Materials, labour, equipment, temporary facilities, site layouts, and documents, names, cost items, job contents, and priority of tasks/procedures [48,88,105,106,110,112,113]
Project attributes and performance	Location, type, contract status, and key performance indicators (KPIs) (e.g., profits, duration, cost per quantity) [29,79,105,107]
Construction techniques	Sensor, server, BIM functionalities [114,115]
Infrastructure disruption information	Disruption types (e.g., utility burst), severity, and affected stakeholders (e.g., road users) [70,72,89,116,117]
Relation names	The names of relations, which do not connect any entities [55,69]

Table 9
Entity extraction methods.

Method types	Methods
Criticality-based	Extract tokens/n-grams as entities based on criticality (i.e., frequency features or information in KBs): • Frequency-based: extract tokens/n-grams if their frequency features (e.g., TF and TF-IDF) exceed a threshold [61,65,106,117]. • KBs-supervised: extract tokens/n-grams if they or their synonyms/related classes exist in KBs [11,57,74,111].
Heuristic rule-based	Extract tokens/n-grams as entities if they match patterns defined in rules [24,55,66,72–74,78–80,113].
ML/DL-based	Extract entities as a named entity recognition (NER) task with ML/DL models. Model inputs are texts and outputs are tags assigned to certain tokens/n-grams for specifying their entity types (e.g., tasks and materials) • ML models: SVM, KNN, DT, LR, NB, and CRF and ensemble models [73,75,81] • DL models: LSTM, GRU [43,77], and Transformer [114] • Combine LSTM with ML models (e.g., CRF), where the former extracts word-level features while the latter capture features of entities tags [77,79,112]

Table 10
Main relation types.

Level	Relation types	Examples
Level-2	Tuple-relations	spatial relations [39,55,83], stakeholders-obligations/rights [52], components-risks-control measures [80], subject-properties-values [82], co-occurrence/sequence of project entities (e.g., components, tasks, and stakeholders) [31,63,118]
	Logical relations	similar-to (synonymy), disjoint-with, different-from [41,84]
Level-3	Belonging relations	is-a, instance-of, part-of/piece-of (meronymy) [53,59]
	Hierarchical relations	sub-class-of, super-class-of [59]
	Constraining relations	concrete-constrains-pouring, power-supply-constrains-crane [68,76]
	Task dependencies	start-to-finish, start-to-start [112]

Table 11
Methods for relation extraction.

Level	Type	Methods
Level-2	Co-occurrence analysis	Extract relations if co-occurrence frequencies of or correlation statistics (e.g., Chi-square) between tokens/n-grams exceeds a threshold [31,63,65,82]
	Association rule mining or unsupervised ML models	Extract sequences or groups of critical tokens/n-grams as tuple-relations based on distance and rule-mining measures [96]
	Heuristic rules	Extract relations if tokens/n-grams (including tuples) and their sequences match patterns defined in rules [39,40,47,52,69,71,83]
Level-3	ML/DL-based	Train ML/DL models which take texts or entity pairs as inputs and semantic-rich relations (e.g., triples) as outputs, e.g., classifying if a relation exists between an entity pair) • Minimal-supervised ML models [53] • LSTM and multi-layer neural networks [112] • KRL models which are enhanced by stacking embeddings of domain classes in model inputs [68]

Section 5 can only contribute to project management if it can be timely delivered to the parties responsible for certain downstream analysis (e.g., identified risk factors are delivered to the safety manager who then can perform risk prediction). However, the construction industry is notorious for inefficient information exchanging which leads to poor collaboration among parties [5]. This section focuses on the second core NLP function that can help address the challenge, i.e., improving information exchanging by organising project documents and assisting information retrieval.

6.1. Document organisation and generation

Most construction documents are organised by metadata (e.g., date and document type). Metadata is abstract and cannot reflect semantic meanings of document contents, making document organisation and retrieval ineffective [46]. Thus, NLP is applied to classify different documents, e.g., claims, PPRs, specifications, web-pages, and change requests. For this purpose, many studies develop ML models based on vectors (e.g., BOW, TF-IDF) of tokens/n-grams/tuples in key positions of documents (e.g., titles and abstracts), while both unsupervised (e.g., K-means, Rocchio, and hierarchical clustering) [28,67,85,93,107,120,121] and supervised models (e.g., SVM, NB, DT, and ensemble classifiers) [85,120,121] are extensively adopted. Classification can be binary (e.g., positive/negative attitudes of PPRs) [96] and multi-labels (e.g., different claims according to reasons) [20,93,104]. Models can be improved again by incorporating semantic features, especially KB classes [67,107]. A few studies classify documents by computing a matching score between documents and document classes with ordered weighted summation, considering frequencies, positions, and expert opinions of tokens/n-grams in documents and the corpus [24,25].

Manually drafting documents (e.g., reports, meeting agendas, RFIs, and change request) is very time consuming, making document auto-generation necessary. Given automatically generated documents usually have standard structures of contents, a common method is to develop templates for different document types [64,111,122]. To further standardise document contents, KBs (e.g., ontologies) can be applied. Specifically, when drafting documents, ontologies can detect critical tokens/n-grams and suggest their ontological and unambiguous forms (e.g., standard class names) [50].

6.2. Document and information retrieval

This task aims to explore file systems and information models (e.g., BIM) to retrieve documents and information (e.g., sentences, provisions, clauses) within documents (similar to question answering), which largely depends on matching query and text contents. The relevant applications and retrieval methods are listed in Tables 12 and 13, respectively.

Moreover, retrieved information and generated documents can be inserted into BIM to enrich its information richness. For instance, Oti et al. [23] and Xie et al. [57] mapped lessons learned of site issues and facility equipment to BIM elements, respectively. BIM elements and texts were vectorized and cross-compared based on their existence in a reference list formed by IFC entities; other studies created templates of project correspondences to record communication texts which were integrated in BIM for timely reuse [78,118].

7. Downstream applications

Sections 5–6 present the direct application of NLP to texts, while this section introduces four main downstream NLP applications identified from the collected articles, all of which depend on the information obtained through information extraction or exchanging.

7.1. Knowledge discovery and formation

Construction management is knowledge-driven. Thus, it is valuable to discover and form knowledge at two levels (Table 14): 1) project situations (similar to automatic summarisation) and 2) general and deeper knowledge of CS.

Table 15 presents main methods for forming knowledge. However, it is necessary to distinguish entity extraction (Section 5.1) and knowledge formation. They overlap in terms of extracting information, but, the former directly works on texts and extracts entities explicitly mentioned, while the latter can recognise implicit knowledge (e.g., hidden topics).

7.2. Compliance checking

NLP-enabled auto-compliance checking is another hot topic. Compliance checking can be realised by straightforward comparison, e.g., checking if required safety information exists in project plans [74,91] and detecting inconsistency of accident types in reports [130]. However, the mainstream practice involves four steps (Fig. 17). Extracted rule patterns usually take the form of tuple-relations. The rule-logic transformation can include 1) standardisation (if applicable) which adopts schemas (e.g., XML) to standardise contents in regulatory documents [76,84]) and 2) applying transforming rules based on the tuple patterns and their syntactic/semantic features. An example is shown in Fig. 17. As-is information to be checked is extracted from digital documents or information systems (e.g., BIM/GIS) with plug-ins. Finally, compliance checking is conducted by filling as-is information in checking rules. Currently, NLP-based compliance checking is mainly employed to assess building design [40], working procedures dependencies [112], and spatial configurations between underground utilities and their surroundings [39,55,83].

Table 12
Main applications of NLP-based document/information retrieval.

Type	Applications
Document retrieval	Share documents among CS companies (e.g., the e-Cognos project) [50,51,60] Search for online articles or technical documents to guide construction practice [67] Identify CAD drawings/documents for design reuse [46,90] Find technical specifications for engineering problems (e.g., concrete cracks) [84,94,123], building maintenance [124], and remedial actions for incidents [101]
Information retrieval within documents	Find applicable regulations, requirements, clauses and criteria of quality, cost, safety [13,18,82,91], useful contents in web-pages [15,104], component information for answering RFIs and project progress/cost/quantity information in BIM [14,31,49,118,122] based on user queries

Table 13
Main methods for document/information retrieval.

Method type	Explanation
Token/n-gram based	Check existence of query tokens/n-grams in key positions of a document (e.g., titles, abstracts, and keywords) [27,84,94,122]. Queries can be expanded using 1) synonyms and KB classes of tokens/n-grams [14,15,64,81], 2) tuple-relations among tokens/n-grams [71]
Statistic-based	Compute and match statistics of tokens/n-grams in queries and documents, including basic frequency statistics [24,25] and more sophisticated ones (e.g., Chi-square) [56]
Rule-based	Evaluate applicability of documents/information based on contexts (e.g., if a document fits certain topics) and application/exception conditions indicated by patterns of tokens/n-grams and distinct words (e.g., 'should' or 'not') [18,82,84]
Distance-based	Compute and match distance between vectors (e.g., TF-IDF, KL-divergence, and Okapi) of documents and queries (vectors are often enhanced with semantic features) [26,46,48,51,103,104,123].
ML-based	Train ML models to determine if a document should be retrieved given text and meta-features of the query [48]
CBR	Retrieve the most similar documents stored in CBR databases [101,124]
Other	Apply path finding (e.g., Dijkstra) to IFC/IFD (tree structures) to retrieve BIM information by mapping extracted entities to IFC/IFD items [49]

Table 14
Summary of knowledge formed.

KB types	Knowledge and examples
Project level	Reasons behind legal issues e.g., unbalanced duty allocation, cost overrun, and damages to products [11] Safety hazard/risk factors/categories (i.e., reasons of accidents), e.g., lack of supervision and illegal operations [20,21,35,43,73,86,95,99,125] Contract/bidding risk factors, e.g., payment methods, quantity clarification, and unclear duties [30,33] Other project risk factors, e.g., underground conditions, environmental impacts, and market uncertainties [41,95,126] Major site issues, quality problems, and defects/deficiencies [28,92,106,127] Critical resources, parties, and tasks/procedures [68,77] Infrastructure failure patterns (i.e., interconnections among infrastructures and stakeholders and how the failure of one infrastructure cause failure of other infrastructures) Roles and responsibilities of project parties [109,128] End user needs/demands on construction projects and technologies [34,97]
Industry level	Attitudes and perceptions of government, project stakeholders, and public against various issues, e.g., environmental impacts and infrastructure resilience [10,38,129] Job demand, categories, and affecting factors behind [32,128] Evolution of technologies (e.g., BIM) [22,45,115]

7.3. Risk prediction

Text information can indicate potential risks which NLP can to-some-extent predict with the help of formed domain knowledge (Section 7.1). Table 16 introduces main risk types and prediction methods.

However, training data can be inadequate for risk prediction, and different models can suit different datasets. As an example to solve this, Li et al. [12] developed four datasets of main safety accidents, where

Table 15
Main methods for domain knowledge formation.

Aspect	Main methods
Topic identification	Key terms analysis, i.e., derive topics based on existence/frequencies of pre-defined key terms in extracted tokens/n-grams [10,11,21,115,116] LDA [11,20,22,45,97,106,129], STM [32] Apply clustering (e.g., K-means, unsupervised seeding/mapping) and partitioning (e.g., multi-dimensional scaling, modularity maximization) to tokens/n-grams or their vectors and then derive topics by manually interpreting the results [28,101,109]
Factor extraction and classification	Identify principle components using PCA [35] Extract topics based on cosine similarity between vectors of tokens/n-grams and pre-defined classes (i.e., topics) [95] Train ML models [37,38,99] and DL models [20,92], taking extracted tokens/n-grams/tuples as inputs and factors as outputs
Relationship exploration	Interpret mutual relations based on co-occurrence frequencies [88,107,109,126], correlation statistics (e.g., Chi-square), and association rule patterns [72,116,118]. The interpretation can be manual or enabled by SVD, PCA, and Analytic Hierarchy Process [36,97,101,128] Network analysis • Node- or edge-level analysis [22,37,66,88,115] • Network- or graph-level analysis [38,102]
Knowledge evolving	Analysing change of knowledge formed using one of above three approaches over time [45,106,115,118]

documents were represented by *doc2vc* vectors. When new accident reports came in, the most similar dataset was selected based on *doc2vec* similarity to share key factors for predicting injury degrees. Then, a meta-leaner was trained to recommend ML model based on dataset meta-features (e.g., the proportion of categorial factors) to maximise prediction performance.

7.4. Other project management functions

Apart from above mainstream topics, Table 17 reveals minority applications of NLP in other project management aspects.

It can be argued that the application of NLP for project management is still limited compared to that of computer vision (CV) [6], but the coverage of NLP in smart construction keeps increasing in the last decade.

8. Discussion

8.1. Insights of applying NLP in smart construction

8.1.1. The role of NLP in the framework of smart construction

Based on insights in [1,5,6], a simplified framework of smart construction is illustrated in Fig. 18, consisting of two pillars and three features. NLP is essential for smart construction, because it has direct and indirect contribution to the digitalisation pillar. Most data in CS is unstructured texts [2,3]; thus, NLP directly and significantly improves effectiveness and efficiency when handling such data using text cleaning and information extraction. NLP also indirectly facilitates AI-enabled decision-making and BIM-based management. According to the decision-making pyramid, decisions are made from data following four steps: data, information, knowledge, and decisions [133], while only three of them are covered by core functions of NLP. Therefore, it is more beneficial that NLP supplements other functions by providing useful data. For instance, optimising infrastructure O&M plans with planning techniques based on damage severity extracted from twitter and enriching BIM for efficient information exchanging [108,126]. The more capable of NLP methods, the more unstructured data are handled, more sophisticated AI techniques are enabled, and more efficient BIM collaboration is applicable, which make CS more digitalised, automated, and connected, respectively.

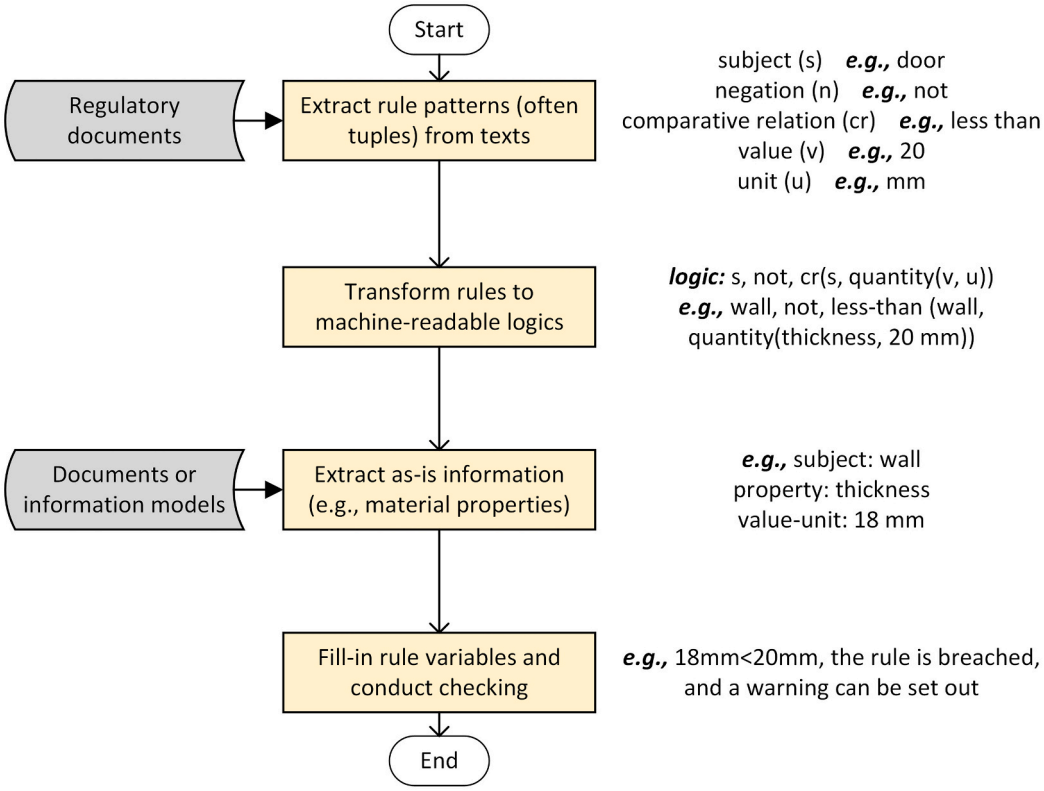


Fig. 17. Steps for auto-compliance checking.

Table 16
Risks and prediction methods.

Risk type	Application	Methods and explanation
Contract	Classify clauses as requirements and non-requirements or distinct categories (e.g., construction, design, and O&M) to prevent ignoring requirements	Train ML models (NB, LR, SVM, and RF) and DL models (CNN and LSTM) [44,131]
	Detect biased/poisonous clauses	• Compare extracted tuple-relation patterns with pre-defined biased patterns [47,52] • Train ML models (DT, NN, and NB) to predict existence of extracted contract risk factors [33]
	Detect substantial contract changes	Train ML models (KNN, SVM, ANN, LR, RF) with texts of change descriptions [19]
	Estimate risk levels	Compute frequency/density of extracted obligation clauses and contract risk factors [18,108] Estimate importance of extracted obligation clauses according to probability of clause breaching and impact (obtained from expert opinions) [13]
	Select bidders	• Train ML models (SVM, KNN, and ANN), based on extracted contract risk factors and common factors (e.g., the number of issued addenda and bidders) [30] • Train DNN models to predict benchmarks of stakeholders' KPI based on extracted KPI entities for tender selection [29]
	Predict litigation results to facilitate dispute resolution	Conduct CBR-based searching [17,54]
Safety	Predict safety risks	Train ML (DNN and ANN) models to identify risk existence based on extracted safety risk factors and then predict possible degree of injury [20,105]
Other	Predict risk levels or classify risk types in other aspects (e.g., environmental, financial, and political)	Train ML models (SVM, KNN, DT, GBRT, NB, and radius-based neighbours (RBN) which is similar to KNN but determines the class based on the number of samples in a fixed radius of the new sample) for risk prediction [41,126]

Table 17
Other NLP-based project management.

Aspect	Application	Methods
Schedule and cost management	Schedule monitoring and deviation detection	Develop a knowledge map of scheduling and cross-compare project situations [127]
	Predict cost overrun (i.e., severe overrun, normal, and underrun)	Train ML models (K-star, RBF NN, and ensemble models) using numerical features (e.g., quantities and schedule) and text data (project descriptions) [16]
	Calculate cost index	• Compare extracted entities in descriptions of pay items to merge inconsistent items • Clean pay items based on statistics (e.g., price ratio) of extracted price information
O&M management	O&M task auto-assignment	Train ML and attention-enhanced DL models, taking O&M documents as inputs and task types and priority as outputs [92,110]
Infrastructure O&M management	Evaluate incident severity, prioritise O&M actions, and provide advice for future infrastructure projects	Sentiment analysis (i.e., positive, neutral, and negative) according to 1) frequencies of pre-defined sentiment words [10], and 2) ML models (SVM, RF, NB, ensemble classifiers) [75,89,132]
Constraint management	Identify unremoved constraints for current tasks	Network analysis by exploring project KGs with path reasoning [2,77]

8.1.2. The expansion of NLP-related application areas

NLP studies present an unbalance of application areas partially due to the supplementary role. Most studies, especially those early ones, focus on document organisation/retrieval [51,90,122]. It is not until 2015 that NLP is increasingly adopted for various engineering applications, because of two reasons, i.e., data availability and algorithm capacity. In the last decade, CV is the common method to support applications mentioned in Section 7. However, most CV models are supervised models, and collecting images for training is time-consuming and expensive (e.g., when high-quality images are needed) [9]. In contrast, collecting texts is much quicker and cheaper. Most text data (e.g., pdf) is accessible from engineers' PC, and more data exists online. Therefore, much data can be collected in a short time (e.g., using crawlers) without using expensive equipment. Besides, texts are easy to be cleaned and often encounter less privacy issues [3].

Besides, text analysing algorithms have achieved significant improvements. Early algorithms depend on sparse BOW/TF-IDF, which cannot capture complex text features to extract entities and relations with rich semantics therefore hinder downstream applications (e.g., knowledge discovery based on extracted entities) [121]. In contrast, word embeddings can effectively capture such features thus increase capacity of ML models. Moreover, word embeddings are complementary to state-of-the-art DL models (e.g., CNN, LSTM, and Transformer), which enable many complex analysing tasks [8]. Nevertheless, as discussed, the expansion of applications is a good sign that the importance of NLP for smart construction is increasing.

8.1.3. The trade-off between efficiency and accuracy

Fully understanding human-written texts is extremely challenging, as texts are very unstructured, complex, ambiguous, and noisy (e.g., missing/incorrect words). Sophisticated text analysis needs a deep understanding of human knowledge, while no current AI can satisfy [7]. DL NLP models are increasingly implemented in CS. However, training DL models requires manually labelling a large amount of data, which is impractical for project teams; besides, DL models do not necessarily reach superior performance over ML models. This is because different models suit different tasks and data, while SVM and ensemble models trained with TF-IDF often achieve the best performance [44,86,120]. Meanwhile, it is time saved and working efficiency increasing that bring substantial benefits in practice. Thus, NLP for smart construction should reach a accuracy-efficiency trade-off, which again reflects its supplementary role. Rather than struggling to increase accuracy by 2% (e.g., 88%–90%) when extracting 50% information at the cost of developing models with excessive data demands and complexity, it is more beneficial to extract 80% information with 80% accuracy, completely saving 80% time and leaving 20% work for human engineers [91]. In fact, experienced engineers are very good at high-level tasks (e.g., detecting contract risks) given key information (e.g., obligations and duties) is extracted from raw texts using NLP [2,7].

8.2. Challenges and future directions

Although NLP can drive smart construction, many challenges exist, which together with potential directions enabled by cutting-edge techniques, are discussed below.

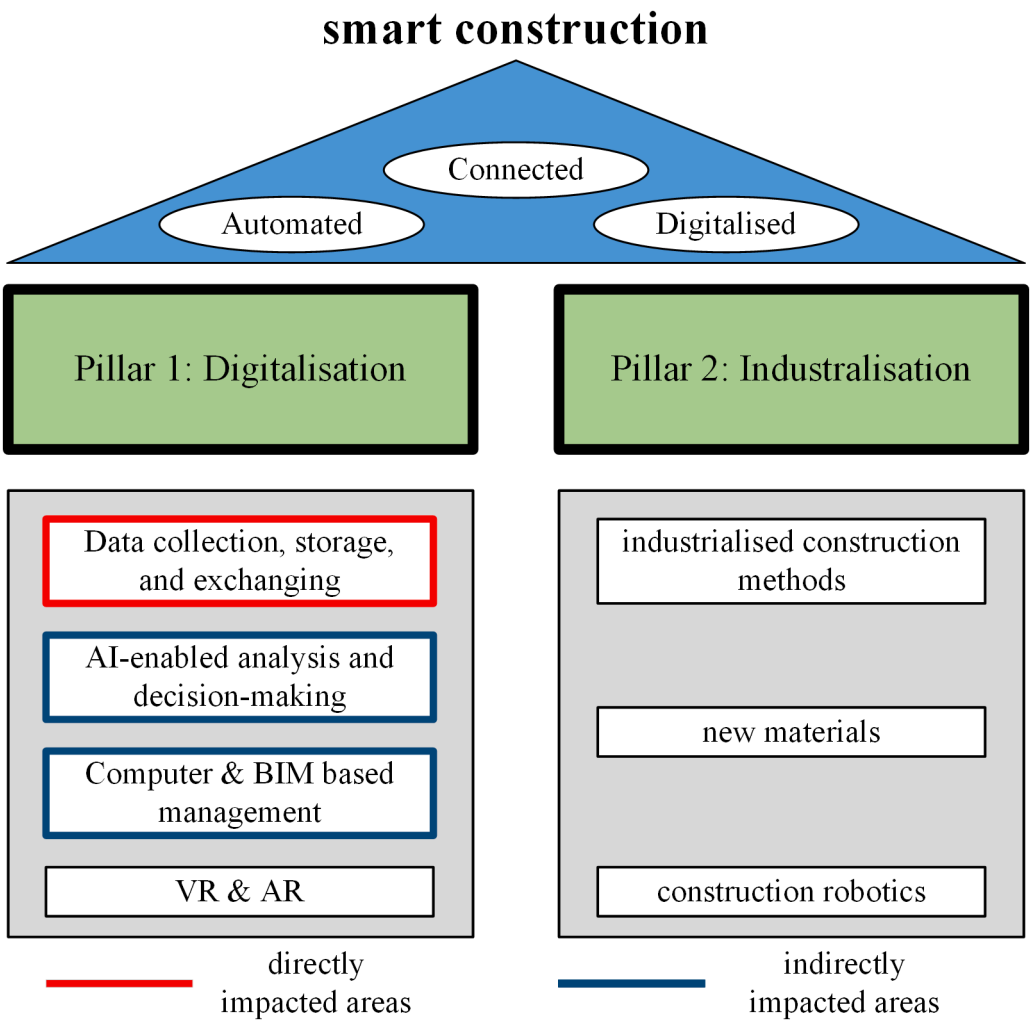


Fig. 18. A simplified smart construction framework.

- **Challenge-1: excessive manual work**

Currently, applying NLP techniques needs much manual work, while the increasing of application areas further worsens the situation. For one thing, rule-based methods need to manually discover text patterns and develop rules; ML-based techniques depend on manual feature engineering and data labelling; DL models, although automate feature extraction, still need to manually label data [134]. In addition, KBs are usually adopted to improve NLP methods, most of which are manually built [2,8,9]. The volume of text data is often much larger than that of images, which is a double-edged sword. Big data improves model capacity but significantly increases manual workload, which necessitates supplementary methods to alleviate the burden.

- **Challenge-2: poor relation extraction**

NLP techniques in CS focus on entity extraction and can extract most entity types [1]. In contrast, relation extraction is less addressed. A few studies apply DL and complex rules to extract semantic-rich relations (e.g., constraining relations among project resources). However, the infeasibility of rules affects their usefulness, while the performance of DL models remains low (around 80%), especially for relations with uncertain directions (e.g., in the construction stage, scaffolding 'constrains' labour, whereas in the temporary facility construction stage, scaffolding and labour 'constrains' each other) [68]. Poor relation extraction restricts NLP applications and increase manual workload, as relations are more valuable for developing KBs, extracting information, and discovering knowledge [134].

- **Challenge-3: poor mapping between functions and NLP techniques**

No superior NLP technique outweighs others. An accuracy-efficiency trade-off exists when applying NLP, where the effectiveness is subject to specific data and functions (i.e., tasks). Unfortunately, few studies can assist selecting the most suitable NLP techniques for specific tasks. This hinders NLP implementation in smart construction, as it is impractical for project teams with inadequate computer background to make such selection. The work of [12] is inspiring for addressing the challenge, which trained meta-learners to first determine appropriate ML models according to features of testing data (e.g., the number of samples and categorial features). However, that study is restricted to safety documents and ML model selection, leaving enormous room for improvements, e.g., automatically selecting rule-based, ML, and DL based techniques by developing more capable meta-learners.

- **Direction-1: Information extraction improvement**

Studies in this direction can address challenges 1–2. KRL models (see Section 5.2) can significantly improve relation extraction. Besides, many variants of KRL models (e.g., TransR, ConvKB, ConvE) are proposed in recent years and are worth being applied in CS to realise more robust

information extraction [134]. Such techniques not only assist extracting semantic-rich entities/relations, but also to-some-extent automate rule design, as many rules are derived from tuple-relations. Furthermore, distant supervision (DS) can supplement KRL models as an effective auto-labelling method to save manual workload. DS assumes that if two entities hold a relation in KBs, all texts (e.g., sentences) containing the entities imply the relation and are labelled as valid triples [135]. Hence, DS can improve KRL models by automatically labelling much more training data. Given the strong assumption causes much noise, the attention mechanism can be adopted to learn probabilities of sentences expressing different relations by simultaneously learning their attention values (i.e., weights) [136].

- **Direction-2: KBs auto-development**

This direction is complementary to direction-1 thereby also addressing challenges 1–2. As mentioned, it is critical to realise auto-development of KBs with rich semantics (e.g., ontologies and KGs). This requires effectively extracting relations as bottom-level components. Although KRL models and DS can improve relation extraction, extracted information often contains many errors (e.g., ambiguous and missing information) hence cannot be used to develop KBs [119]. Two methods based on KBs' topology can be used to tackle the problem. One is rule-based, which adopts semantic reasoning for disambiguation (e.g., inferring two entities as synonyms) and detecting missing entities/relations (e.g., if certain nodes are linked to a central node, then a link should be established between the central and another node). The other is based on the graph neural network (GNN). GNN represents cutting-edge DL models for KBs completion. It can automatically recognise topology patterns defined in above reasoning rules and then classify duplicated nodes and predict missing nodes/edges by integrating information in a neighbourhood of node/edges, which has been adopted for financial fraud detection and transportation flow prediction [137].

- **Direction-3: multi-modal information integration**

Studies in this direction can address challenges 2–3. Current studies focus on text data while ignore other information modals apart from KBs, which can supplement NLP, e.g., images and audios. Information of some entities can be ambiguous when being described only by texts. For instance, descriptions of similar materials contain many similar tokens/n-grams; therefore, ML/DL models require more training data for extracting them but still suffer low accuracy [9]. The models can be enhanced by incorporating images if it is easier to distinguish these entities in images. Specifically, CNN can extract image features as vectors and concatenate them to word embeddings for training multi-modal models [137]. Moreover, when text data of certain entities/relations is inadequate, images can serve as latent information to realise zero- or few-shot learning [137].

Typing texts can be inefficient especially for onsite employees. In this case, utilising audio data is a useful alternative. Texts and audios are both human languages, and audios are often converted to texts before further analysis. This involves additional steps of automatic speech recognition (ASR), i.e., signal processing and pattern recognition. Relevant studies in CS are inadequate and limited to audio-based BIM information retrieval using mature audio-to-text tools [14]; thus, more studies are needed to explore the potential of integrating audio data in other NLP functions (e.g., audio-triggered event extraction from onsite reports).

- **Direction-4: framework development for NLP-enabled decision-making**

This direction mainly addresses challenge 3. With the aim to maximise management and decision-making efficiency by applying NLP to texts, a comprehensive framework can be developed. At the bottom of the framework lie NLP techniques and NLP-supported functions/tasks. Sections 4–7 provide a good starting point but need elaboration. Then, indicators of applying NLP techniques should be proposed considering both accuracy and efficiency, including those introduced in Section 4.5 and more efficiency measures, e.g., data demands, training time, and model complexity. Finally, a detailed function-technique mapping is set up. This can help evaluate suitability of NLP techniques with the indicators and specific tasks and data. For instance, with the mapping results, an engineer can identify certain entity/relation types with distinct patterns and train ML models as the cost-effective method to extract them; while it is more efficient to extract remaining entities/relations using rules or manual approaches. Moreover, when more data (i.e., mapping results) is available, a meta-leaner which is the expanded version of [12], can be trained to assist selecting NLP-techniques automatically thereby increasing their practicality. A summary of above discussion is shown in Fig. 19.

- **Future NLP-based implementation areas**

With the development of NLP, current implementation scenarios will be improved, and more will be enabled. Specifically, enhanced information filtering/extraction will be able to extract both critical entities and relations. This will further reduce workload of engineers as they do not need to manually investigate relations among entities. Document organisation and information retrieval systems will be more accurate and comprehensive. They can handle more complex documents (e.g., freely written narratives) rather than merely those with structured contents (e.g., contracts). They can understand queries written by natural languages, which shields engineers from the difficulty to learn query languages. Future KBs or expert systems will cover more backgrounds and answer more complex questions, as they will be equipped with powerful semantic reasoning and can continuously enrich themselves using both online and project information through auto-KBs development methods. They also support natural language or even audio queries. Automated compliance checking tools will be more flexible, which will not be restricted to certain areas as rule-based checking does while keeping high accuracy and efficiency. Moreover, NLP will enable/improve more downstream applications. For instance, network analysis tools enabled by relation extraction can facilitate management at project-level (e.g., finding critical constraints) and city-level (e.g., identifying vulnerable infrastructures). Besides, current project management models predicting project performance and risks rely on certain fixed features (e.g., project type and bidders' revenue). NLP will contribute to the models by feeding them with rich information extracted from text data (e.g., project descriptions and comments from stakeholders).

9. Conclusion

In the last decades, an increasing number of studies are conducted to solve problems in CS through NLP. This paper provides a state-of-the-art review appraising studies and applications of NLP in CS. In summary, the core functions of NLP, i.e., information extraction and document organisation have been largely improved with various techniques. NLP can also assist many downstream applications and enhance decision-making and management effectiveness. However, additional efforts are needed to continuously improve relation extraction, reduce manual workload, and develop a framework for NLP application considering the accuracy-efficiency trade-off. To the authors' best knowledge, the paper is unique as: unlike other efforts reviewing AI in CS, which only generally introduce NLP, this paper is the first that provides 1) comprehensive introduction of bottom-level techniques (Section 4), core applications (Section 5–6), and downstream applications (Section 7), and 2) in-depth discussion of insights, challenges, and future directions and implementations. These findings serve as valuable agenda for leveraging NLP to realise smart construction. The study also has

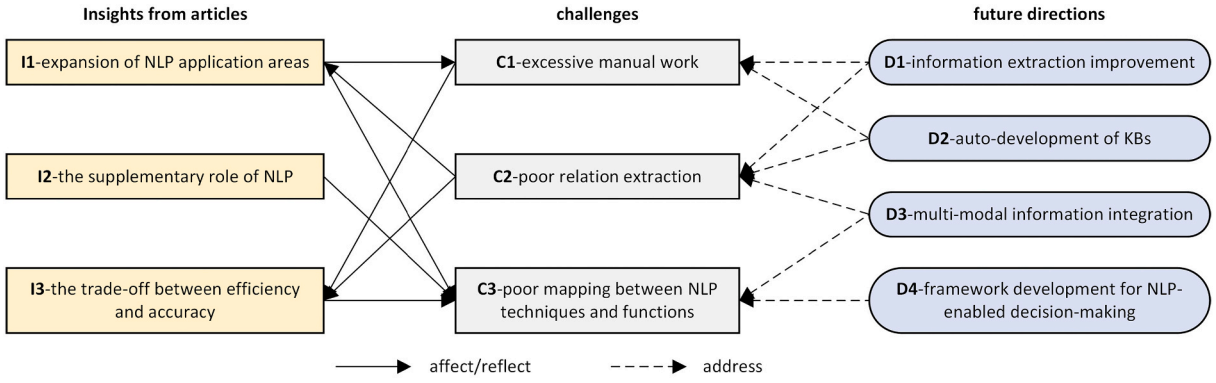


Fig. 19. Summary of discussion

practical value, which provides project teams useful references to recognise cutting-edge NLP techniques and to-some-extent spot suitable ones according to specific needs.

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Declaration of Competing Interest

None.

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